



LungLens

See what the AI sees. Explainable chest X-ray screening for earlier pneumonia detection.

UnivaBio Hackathon 2026
Theme: AI for Human Health

The problem

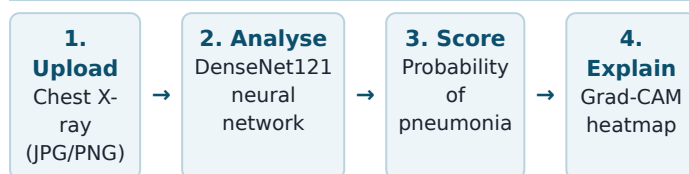
Pneumonia is the single biggest infectious killer of young children: the WHO reports it caused 14% of all deaths of children under five in 2019 (about 740,000 children). A chest X-ray is the standard way to confirm it, but many hospitals and clinics do not have a radiologist available to read X-rays quickly, so diagnosis and treatment are delayed.

AI can read X-rays in seconds, but most AI tools are **black boxes**: they give an answer without a reason. Doctors are right not to trust an answer they cannot check.

Our solution

LungLens is a web app that screens a chest X-ray for pneumonia in a few seconds and **shows its reasoning**: a heatmap highlights the parts of the lungs that drove the decision, so a doctor can see at a glance whether the AI is looking at real disease or at something irrelevant.

How it works

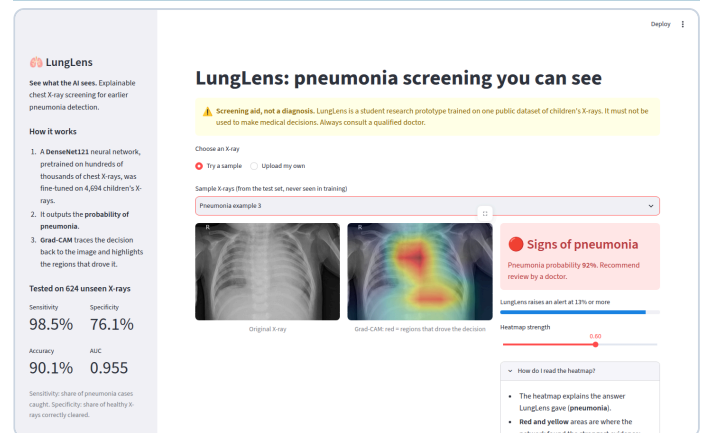


- **Transfer learning**: we started from a DenseNet121 already trained by the open-source TorchXRyVision project on hundreds of thousands of adult chest X-rays, then fine-tuned it on 4,694 pediatric X-rays (Kermay et al., 2018).
- **Fair evaluation**: validation images were split *by patient*, so the model is never tested on a child it has already seen. The final test set of 624 X-rays was used only once, at the end.
- **Balanced decisions**: the dataset has 3x more pneumonia than normal images, so we weighted the training loss and chose the alert threshold that balances sensitivity and specificity.
- **Grad-CAM** (Selvaraju et al., 2017) uses the network's own gradients to measure how much each region of the image contributed to the answer.

Key features

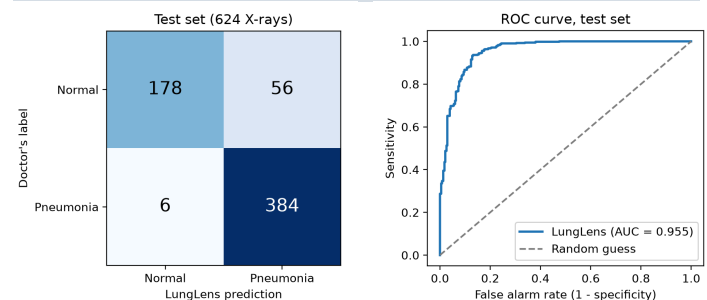
- Upload your own X-ray or try built-in samples from the test set.
- Clear result: pneumonia probability, alert threshold and a plain-language verdict.
- Side-by-side original X-ray and adjustable heatmap overlay.
- Built-in guide to reading the heatmap and spotting model shortcuts.
- Sanity check that warns when an upload does not look like an X-ray.

See it in action



Results on 624 unseen test X-rays

Accuracy	90.1%	AUC (1.0 = perfect)	0.955
Sensitivity (sick caught)	98.5%	Specificity (healthy cleared)	76.1%
Precision	87.3%	Missed pneumonia cases	6 of 390



Responsible AI

Screening aid, not a diagnosis. LungLens was trained on children aged 1 to 5 from one hospital, so it may not work as well on adults or on X-rays from other machines. It flags X-rays for a doctor to review and never replaces one. The heatmap exists so that people can question the AI rather than blindly trust it.

Built with

Python · PyTorch · TorchXRyVision · scikit-learn · Streamlit · Matplotlib. Data: Kermay et al., *Cell* 2018 (CC BY 4.0). Trained on an ordinary 4-core CPU in under 90 minutes.